INCH-POUND

MIL-P-71093 (AR)
26 August 1992

MILITARY SPECIFICATION

POWDER, IGNITION, GASLESS, F-33B

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for composition F-33B for the procurement of gasless, ignition powder (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-C-96	Cans, Metal, 28 Gage and Lighter
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Locker corner

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 1375
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MILITARY

MIL-Z-399	Zirconium (Granular and Powdered)
MIL-E-463	Ethyl Alcohol (For Ordnance Use)
MIL-I-706	Iron Oxide, Ferric Red Dry (Natural and Synthetic)
MIL-D-20550	Diatomaceous Earth
MIL-V-23776	Vermiculite, Expanded
MIL-I-24391	Insulation Tape, Electrical, Plastic Pressure Sensitive
MIL-B-26701	Bottles, Screw Cap and Carboys, Polyethylene Plastic

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-1234	Pyrotechnics, Sampling, Inspection and Testing

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS

49 CFR 170-190 -	Department of Transportation Rules and Regulations for the Transportation of Explosives and other Dangerous Articles

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AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D2905 - Sampling Industrial Chemicals

The Interstate Commerce Commission Regulations are now a part of The Code of Federal Regulations, available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders for the above publication should cite, "49 CFR 170-190" (latest revision).)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. The materials used in the manufacture of the gasless ignition powder F-33B shall conform to the following.

3.1.1 Zirconium powder. The zirconium powder shall meet the requirements of MIL-Z-399 type II, class 1.

3.1.2 Ferric oxide powder. The ferric oxide powder shall meet the requirements of MIL-I-706, type I, class 2, with the following additions.

3.1.2.1 Total iron. Total iron shall be 69.7 $\pm$ 0.5 percent when determined in accordance with 4.5.1.1.

3.1.2.2 Average particle size. Average particle size of ferric oxide powder shall be less than 1 micron when determined in accordance with 4.5.1.2.

3.1.2.3 Total oxygen content. Total oxygen content shall be 29.8 percent minimum when determined in accordance with 4.5.1.3.

3.1.3 Diatomaceous earth. The diatomaceous earth shall meet the requirements of MIL-D-20550 with the following additions.

3.1.3.1 Weight loss on ignition. Weight loss of the diatomaceous earth on ignition shall be a maximum of 0.4 percent when determined in accordance with 4.5.2.1.

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3.1.3.2 Silica content. Silica content shall be a minimum of 90.0 percent when determined in accordance with 4.5.2.2.

3.1.3.3 Granulation. Material is acceptable if 99.0 percent will pass through a U.S standard sieve 325 when tested in accordance with 4.5.2.3.

3.1.4 Ethyl alcohol. The ethyl alcohol shall meet the requirements of MIL-E-463.

3.2 Composition. The percentage of composition by weight of the F-33B powder shall be in accordance with table I. Acceptance of the powder under the test of 4.5.4 shall be considered in satisfactory compliance with the composition formulation.

3.2.1 Formulation procedure. The ingredients in the proportions by weight given in table I shall be mixed in ethyl alcohol (100 milliliters per 100 grams dry mix). The mixture shall be subdivided into approximately 250-gram increments and packaged. An advisory manufacturing procedure which has been found to produce satisfactory material is given in 6.4.

TABLE I. Composition of F-33B gasless ignition powder.

Ingredient	Composition (% by weight)
Zirconium powder	41.0 ± 1.0
Ferric oxide	49.0 ± 1.0
Diatomaceous earth	10.0 ± 1.0

3.2.2 Change of qualified formulation. After qualification, the manufacturer shall notify the purchaser prior to the implementation of any change in the formulation already established by paragraph 3.2.1. Such notification shall include a thorough description of the proposed formulation and a suggested plan designed to demonstrate that the new formulation will not adversely affect performance, quantity, reliability or interchangeability and that the changed product will continue to meet the specification requirements. The purchaser will review the proposed change and test plan and advise the manufacturer of the testing required to obtain approval of the proposed change. Upon completion of the prescribed tests, the results shall be submitted to the purchaser for review. Based upon the review, the purchaser will approve or disapprove the change and notify the manufacturer. At the manufacturer's option, devices incorporating

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the change may be manufactured and tested at the manufacturer's expense prior to approval; however, all shipments of these changed devices shall be withheld until approval is granted by the purchaser.

3.2.3 Moisture content of F-33B. The moisture content of the F-33B gasless ignition powder after drying and screening shall not exceed 0.05 percent by weight when determined in accordance with 4.5.3.

3.2.4 Heat output. The heat evolved by the F-33B gasless ignition powder when burned under an argon atmosphere of 25 atmospheres initial pressure shall be 435 to 475 calories/gram when determined in accordance with 4.5.4.

3.3 Workmanship. The mixing of the ingredients in each batch and the blending of the batches of ignition powders shall be of such quality as to produce a homogeneous lot free from lumps and foreign materials.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to accept defective material.

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4.2 Sampling.

4.2.1 Lot. A lot of F-33B gasless ignition powder shall be limited to a batch produced in one mixing or blending operation. Maximum lot size shall be 50 pounds.

4.2.2 Sample. The F-33 powder composition shall be divided into five containers. Five samples shall be taken from each container and their test results shall be averaged to meet the requirement of 3.2.4. Samples shall be identified in accordance with 5.3.1.

4.2.2.1 Alternate Sample. Individual samples shall be selected from 10 percent of the bottles after packing or a minimum of two samples, whichever is greater. Heat output tests shall be performed on each sample and their test results averaged to meet the requirement of 3.2.4. All other tests shall be conducted on a composite of these samples. Samples shall be identified in accordance with 5.3.1.

4.3 Test equipment and materials. In addition to standard laboratory equipment, the following special equipment will be required to perform the acceptance tests set forth in this specification.

(a) Automatic Calorimeter - An automatic calorimeter with thermometers, rotor stirrer drive, water cans and bomb (see 6.3).

(b) Benzoic acid pellets for standardizing oxygen bomb calorimeter - standard grade, prepared by National Bureau of Standard; Washington, D.C. These pellets are recommended by Parr Instrument Company Inc. since they undergo complete combustion. Powdered benzoic acid is not recommended. Instructions for standardizing the calorimeter are given in the manufacturer's operating manuals (see 6.3).

4.4 Test conditions. Unless otherwise specified, tests shall be conducted at a temperature of 25 ± 5 degrees C.

4.5 Methods of inspection. All tests given in this section shall be performed using prescribed procedures for replicate determinations given in standard analytical chemistry textbooks or ASTM D2905. See 6.6 for use of equivalent test methods.

4.5.1 Ferric oxide.

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4.5.1.1 Total iron determination. Take an approximate 2-gram sample from each container of the ferric oxide powder to be used in making the F-33B gasless ignition powder and dry at 110 degrees C until successive weights at 30-minute intervals agree to within ± 0.002 gram. Weigh 0.2 gram ± 0.002 gram of dried sample and place in a 250-milliliter beaker. Add 10 milliliters of 1:1 hydrochloric acid by volume. Cover the beaker with a watch glass and warm gently until all the sample is dissolved. Dilute with distilled water to about 50 milliliters. Pass the solution through a Jones reducer (or equivalent). The effluent from reductor is best collected under nitrogen. Wash t reductor with se of water. Titrate the reduced iron in the filtering flask with standardized 0.1N potassium dichromate ($K_2Cr_2O_7$) solution using 15 ml of the indicator solution. Titrate to the appearance of a permanent blue-purple color. Record the number of milliliters of $K_2Cr_2O_7$ solution required. Calculate the percent iron as follows:

$$\text{Percent Fe} = \frac{A \times N \times 0.05585}{SW} \times 100$$

Where:

A = milliliters of potassium dichromate solution required.
N = normality of potassium dichromate solution
SW = grams of sample weight

If the sample fails to meet the total iron requirement of 3.1.2.1, the lot shall be rejected.

4.5.1.1.1 Indicator solution. Weigh 0.095 gram of barium diphenylamine sulfonate and add to 100 milliliters of water. Stir until solution is complete. Add 10 milliliters of dilute sulfuric acid (1:1). Let stand several hours and filter. Add 500 milliliters of 85 percent phosphoric acid and dilute to 1 liter.

4.5.1.2 Average particle size. The laboratory test for determining the average particle size of the ferric oxide powder shall be made in accordance with MIL-STD-1234, method 202.1. If the sample fails to meet the average particle size requirement of 3.1.2.2, the lot shall be rejected.

4.5.1.3 Total oxygen determination. Take an approximate 2 gram representative sample from each container of the iron oxide powder to be used in making tition powder.
Dry the sample at 100 degrees C until successive weighing at 30 minute intervals agree to within ± 0.002 gram. Weigh

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approximately 0.25 gram with an accuracy of ± 0.2 milligrams of the dried iron oxide (Fe_2O_3) powder into a combustion train through which dry, oxygen-free hydrogen can be passed. Heat the tube by placing it in an electrically heated combustion furnace, gradually bring the temperature to a dull red heat. The water evolved from the reduction of the Fe_2O_3 is collected in a tared tube containing a magnesium perchlorate water absorbent. After all moisture is swept into the absorbent, weigh the tube to obtain the weight of the water absorbed. Calculate the percent oxygen as follows:

$$\text{Percent } O_2 = \frac{\text{Weight of water absorbed}}{\text{sample weight}} \times 88.88$$

If the sample fails to meet the total oxygen requirement of 3.1.2.3, the lot shall be rejected.

Note: Exact weight of tube and $Mg(ClO_4)_2$ must be recorded prior to experiment.

4.5.2 Diatomaceous earth.

4.5.2.1 Weight loss on ignition. Take an approximate 2-gram sample from each container of diatomaceous earth to be used in making the F-33B gasless ignition powder. Dry the sample to 110 degrees C until successive weighing at 30 minute intervals agree to within ± 0.002 gram. Weigh 0.15 gram ± 0.2 milligram of dry sample into a tared 30-milliliter platinum crucible and cover sample with a platinum lid. Heat carefully over a Meker burner (or equivalent) to about 1100 degrees C for 10 minutes. Cool the crucible in a desiccator and weigh. Calculate the percentage of weight loss based on the sample weight as the percentage of weight loss on ignition. If the diatomaceous earth fails to meet the weight loss on ignition requirement of 3.1.3.1, the lot shall be rejected.

4.5.2.2 Silica content determination. Moisten the ignited material in the crucible with a few drops of water. Add 10 drops of 1:1 sulfuric acid. Fill the crucible about one-half full of 48 percent to 51 percent hydrofluoric acid and place on a sand bed on a hot plate in a hood. Heat until copious fumes of sulfur trioxide are evolved. Continue heating at higher heat until the residue appears dry. Then heat carefully over a flame until no further fumes are evolved. Heat the crucible to 1100 degrees C for 10 minutes. Cool and weigh the crucible. Calculate the difference in determination as the percentage of silica. If sample fails to meet the silica content requirement of 3.1.3.2 the lot shall be rejected.

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4.5.2.3 Granulation. Take a 100-gram sample from each container of diatomaceous earth to be used in making the gasless ignition powder. Transfer the sample to a U.S. Standard Sieve 325. Wash with a spray of water until no more material passes through. Recover the material retained on the screen by washing it into a beaker and filtering it through a tared filtering crucible of fine porosity. Wash the collected material several times with alcohol and dry at 100 degrees C. Weigh the material and calculate the percentage passed through the sieve. If sample fails to meet the granulation requirement of 3.1.3.3, the lot shall be rejected.

4.5.3 Moisture content of F-33B gasless ignition powder. Transfer approximately 2 grams of gasless ignition powder from the representative sample of 4.2 to a tared aluminum weighing dish fitted with an aluminum lid. Weigh to the nearest 0.1 milligram and calculate weight of sample. Transfer weighing dish and sample to a vacuum drying oven set at 71 ± 2 degrees C. Remove lid and dry sample under vacuum (28 inches Hg) until constant weight is obtained. Cool and weigh after drying. Calculate the weight loss as percentage of moisture in sample. Weighing shall be accurate to ± 0.2 milligram. If the sample fails to meet the moisture requirement of 3.2.3, the lot shall be rejected.

4.5.4 Heat output.

4.5.4.1 Apparatus. The test apparatus shall consist of a calorimeter complete with thermometer, stirrer motor drive, ignition unit, and an adapter for pressuring gas. (See 6.3).

4.5.4.2 Equipment and accessories. In addition to the apparatus described above, the following shall be required:

(a) Argon gas supply in pressure cylinder

(b) Standard benzoic acid pellets for standardizing calorimeter.

4.5.4.3 Procedure. Standardize the calorimeter in accordance with the manufacturer's manual of instructions. Weigh a sample of appropriate size for the instrumentation being used and place in the bomb following manufacturer's instructions. Purge the bomb with argon to remove all air then pressurize with 25 atm of argon and determine the heat output in accordance with manufacturer's instructions. If the sample fails to meet the heat output requirement of 3.2.4, the lot shall be rejected.

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5. PACKAGING

5.1 Preservation and packaging.

5.1.1 Level A. In addition to the following, Level A packaging shall be in accordance with 49 CFR 170-190.

5.1.1.1 Unit packaging. A mixture of approximately 250 grams maximum of gasless ignition powder and a minimum of 25 percent alcohol conforming to MIL-E-463 shall be packaged in a polyethylene bottle conforming to the requirements of MIL-B-26701. Inside diameter at the screw cap end of the bottle shall be a minimum of 1.0 inch. The bottle shall be sealed and secured in place with not less than three turns of electrical tape conforming to MIL-I-24391 around the stopper and not less than two turns of tape around the length of the bottle.

5.1.1.2 Intermediate packaging. Unit containers shall be intermediately packaged in a container conforming to the requirements of PPP-C-96, type V - class 2, or type IV, with slip cover provided with leak-proof seal and locking tabs. The cover shall be held in place by soldering or crimping in at least four points. Size I, II, or IV expanded vermiculite material conforming to the requirements of MIL-V-23776 shall be used in the intermediate container to provide insulation shielding and to insure a tight pack.

5.2 Packing. Packing shall be level A or B as specified in the contract or purchase order.

5.2.1 Level A. In addition to the following, level A packing shall be in accordance with 49 CFR 170-190.

5.2.1.1 Exterior containers. Intermediate containers shall be packed in an overseas-type, wood-cleated plywood box conforming to PPP-B-601 or in a class 2, style 2, 2.5, 3, or 4, nailed wood box conforming to PPP-B-621. Vermiculite material, as specified in 5.1.1.2, shall be used within the selected shipping box to separate and provide insulation shielding of the intermediate containers and to ensure a tight pack. The gross weight of a given shipping container shall not exceed 100 pounds.

5.2.2 Level B. In addition to the following, Level B packing shall be in accordance with 49 CFR 170-190.

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5.2.2.1 Exterior containers. Intermediate containers shall be packed in a domestic-type, wood-cleated plywood box conforming to PPP-B-601 or in a class 1 style 2, 2.5, 3, or 4, nailed wood box conforming to PPP-B-621. Vermiculite material, as specified in 5.1.1.2, shall be used within the selected shipping box to separate and provide insulation shielding of the intermediate containers and to ensure a tight pack. The gross weight of a given shipping container shall not exceed 100 pounds.

5.3 Marking.

5.3.1 Identification. Each lot shall be identified by:

(a) Lot number
(b) Manufacturer
(c) Date of manufacture

5.3.2 Special markings. Special marking requirements shall be in accordance with 49 CFR 170-190.

5.3.3 Normal markings. In addition to any special markings required by the contract or order, all markings shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The gasless ignition powder is used to transfer ignition from primers to gasless delays and from delays to detonators or propelling charges.

6.2 Acquisition requirements. Procurement documents should specify the following:

(a) Title, number, and date of this specification

(b) Type, style, material class, grade, and heat treatment or condition of fasteners (see 3.1, 3.2, and table I).

(c) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1)

(d) Level of preservation and packing (see 5.1 and 5.2)

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(e) That the safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material" DoD 4145.26M are applicable.
Note: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition Ashore", OP 5, are applicable.

6.3 Test apparatus. Calorimeter, Serial No. 1411, complete with thermometers, motor stirrer drive, and water cans, and oxygen bomb, fabricated by the Parr Instrument Company, Inc., 211 53rd Street, Moline, Illinois 61625, or equivalent may be used for this test. The test apparatus and method of operation is described in Manual No. 128, Instructions for the No. 1411 Combustion Calorimeter, which is furnished with the calorimeter.

6.4 Advisory manufacturing procedures.

6.4.1 Processing of zirconium. The zirconium powder is stored and shipped in water. Slurry zirconium in container by rolling mill until all zirconium particles are in suspension. Filter the zirconium slurry using a filter flask, Buchner funnel, and No. 2 Whatman Filter paper or equivalent. Rinse filter cake twice with alcohol. When filter cake appears dry transfer to tared aluminum container. Immediately take random samples from the cake to make 50 grams. Dry the composite sample in a suitable oven at a temperature not to exceed 60 degrees C (140 F) until two successive weighings agree within 0.02 percent. Calculate percent of zirconium from dry weight divided by wet weight of composite sample. Immediately weigh damp filter cake. Transfer weighed filter cake to container and slurry with alcohol on a jar roller mill. Calculate dry weight of zirconium which includes total of dry sample weights, and dry filter cake weight. Other ingredient weights are calculated on the basis of the calculated dry weight of zirconium.

CAUTION: The operator should wear face shield, fire resistant clothing, asbestos gloves, safety glasses, and safety shoes. Operations such as weighing, filtering, and transferring the dry powder should be done behind a shield. Spatulas and containers should be nonsparking material and grounded.

6.4.2 Drying and storage of ferric oxide and diatomaceous earth. The ferric oxide powder and diatomaceous earth shall be dried separately at 190 $\pm$ 5 degrees F, (85 to 90 degrees C) in a suitable oven until two successive weighings at 1 hour intervals agree to within 0.5 percent. The dried ferric oxide powder and the diatomaceous earth shall be in a tightly closed container or in a suitable oven until ready for use.

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6.4.3 Mixing of gasless ignition powder. The mixing procedure is based on a single batch of ignition powder. The operator must exercise care to maintain safety and reproducibility of mixing conditions to provide a homogeneous mixture of the desired composition of ignition powder. The mixer should be located in an isolated and barricaded area. Weighing and mixing should be performed in a shielded area.

(1) Add ethyl alcohol to a clean mixing bowl which is adapted to the mixer.
(2) Weigh diatomaceous earth and ferric oxide and add to the bowl.
(3) Lower mixing blade (fluid shear type) into bowl and allow to run.
(4) Stop mixer periodically and, using ethyl alcohol, wash accumulated material back into mixing area.
(5) Add zirconium slurry and repeat steps 3 and 4.
(6) Transfer mix into polyethylene bottles (a maximum of 250 grams per bottle). Seal bottles in accordance with 5.1.1.1.

6.4.4 Drying of ignition powder.

(1) Pour material to be dried into a metal drying tray.
(2) Dry at 60 degrees C for 16 hours minimum.
(3) After drying, cool powder for a minimum of 20 minutes.
(4) Transfer the dried and cooled ignition powder to a 30-mesh sieve with porcelain balls or rubber stoppers.
(5) Screen the powder on a Ro-Tap or shaking apparatus operated by remote control and located in an isolated and barricaded area.
(6) When screened, remove the sieve pan behind a shield.
(7) Transfer the screened ignition powder to conductive rubber containers. The maximum quantity placed in each container shall be limited to 50 grams.

6.4.5 Storage of mixed ignition powder. When ignition powder manufactured and stored is over 200 grams, the powder shall be stored wet in well-stoppered polyethylene container. Minimum weight of ethyl alcohol (conforming to MIL-E-463) added shall be 25 percent of weight of ignition powder.

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6.5 Submission of inspection equipment designs for approval. Submit equipment designs, as required to Commander, ARDEC, ATTN: SMCAR-QAR-R, Picatinny Arsenal, New Jersey 07806-5000. This address will be specified on the contract Data Requirements List, DD form 1423 in the contract.

6.6 Equivalent test methods. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted thru the Contracting Officer to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-QAR-R, Picatinny Arsenal, New Jersey 07806-5000. This description should include but not be limited to the procedures used, the accuracy and precision and drawings of any special equipment required.

6.7 Screening for specific defects. At the direction of the contracting officer, and with the recommendation of the preparing activity (SMCAR-QAR-R), batches may be screened for specific defects.

6.8 Subject term (keyword) listing.

Ammunition
Projectile
Delay Element
Pyrotechnic

Custodians:
Army-AR

Preparing activity:
Army-AR

(Project: 1375-A479)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-71093	26 AUGUST 1992

3. DOCUMENT TITLE: POWDER, IGNITION, GASLESS, F-33B

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial 201-724-6675	(2) AUTOVON DSN-880-6675
c. ADDRESS (Include Zip Code) ATTN: SMACA-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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